

## 10. Moments of Inertia

Tuesday, November 12, 2024 9:35 AM

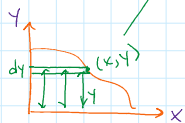
### \* Moment of Inertia

- ↳ has no physical meaning
- ↳ Encountered multiple times in formulas used in fluid mechanics, mechanics of materials, structural mechanics & mechanical design.

### \* Mass Moment of Inertia

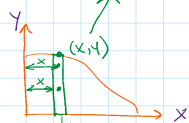
- \* Measure of a body's resistance to angular acceleration.

$$I_x = \int_A y^2 dA$$



\* draw horizontal rectangle

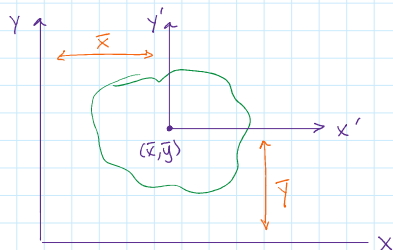
$$I_y = \int_A x^2 dA$$



\* draw vertical rectangle

### \* Parallel-Axis Theorem

- \* Used to get moment of inertia of a body about any axis. The axis is parallel to an axis that passes through the centroid & about which the moment of inertia is known.



$$I_x = I_{x'} + A\bar{y}^2 \rightarrow I_x > I_{x'}$$

$$I_y = I_{y'} + A\bar{x}^2 \rightarrow I_y > I_{y'}$$

\*  $I_{x'}$  &  $I_{y'}$  are typically known (tabulated)

\* Moment of Inertia of a body increases as the body moves away from the axis.

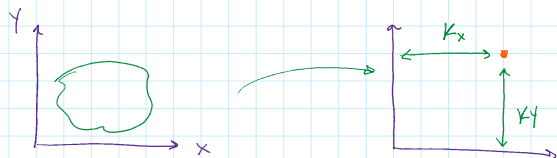
### \* Radius of Gyration

- \* Distance of a point from an axis, such that if the area was concentrated at that point, the resulting moment of inertia would be equivalent to the moment of inertia produced by the distributed area.

$$K_x = \sqrt{\frac{I_x}{A}}$$

$$K_y = \sqrt{\frac{I_y}{A}}$$

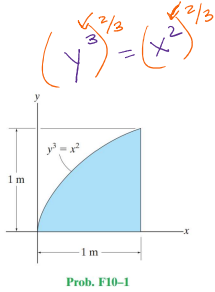
$K$  - radius of gyration.



$$I_x = K_x^2 A$$

$$I_y = K_y^2 A$$

**F10-1.** Determine the moment of inertia of the shaded area about the  $x$  axis.



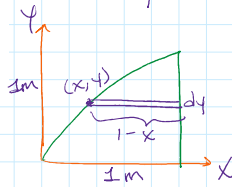
$$I_x = \int_A y^2 dA \quad I_y = \int_A x^2 dA$$

$$y^2 = f(x) = ?$$

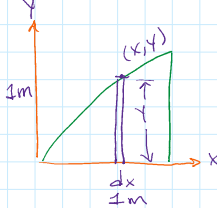
$$y^2 = x^{4/3} \rightarrow y = x^{2/3}$$

$$x^2 = f(y) = ?$$

$$x^2 = y^3 \rightarrow x = y^{3/2}$$



$$dA = (1-x) dy$$



$$dA = y dx$$

$$I_x = \int_A y^2 dA = \int_0^1 y^2 (1-x) dy$$

$$I_x = \int_0^1 y^2 (1-y^{3/2}) dy$$

$$I_x = \int_0^1 y^2 - y^{7/2} dy$$

$$I_x = \left[ \frac{1}{3} y^3 - \frac{2}{9} y^{9/2} \right]_0^1$$

$$I_x = \frac{1}{3} (1)^3 - \frac{2}{9} (1)^{9/2} - 0$$

$$I_x = \frac{1}{3} - \frac{2}{9} = \frac{1}{9} \text{ m}^4$$

$$I_y = \int_A x^2 dA = \int_0^1 x^2 \cdot y dx$$

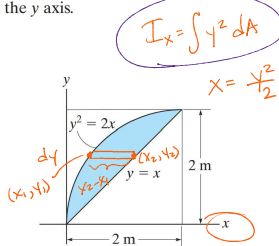
$$I_y = \int_0^1 x^2 \cdot x^{2/3} dx$$

$$I_y = \int_0^1 x^{8/3} dx = \left[ \frac{3}{11} x^{11/3} \right]_0^1$$

$$I_y = \frac{3}{11} [1^{11/3} - 0] = \frac{3}{11} \text{ m}^4$$

10-21 -  $I_x$

**10-22.** Determine the moment of inertia of the shaded area about the  $y$  axis.



$$I_x = \int y^2 dA$$

$$x = \frac{y^2}{2}$$

\* Parallel  $x$ -axis  
horizontal rectangle

$$dA = (x_2 - x_1) dy$$

$$dA = (1 - \frac{y^2}{2}) dy$$

$$I_x = \int_0^2 y^2 (1 - \frac{y^2}{2}) dy$$

$$I_x = \int_0^2 y^3 - \frac{y^4}{2} dy$$

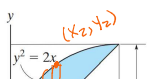
$$I_x = \left[ \frac{1}{4} y^4 - \frac{y^5}{10} \right]_0^2$$

$$I_x = \frac{1}{4} (2)^4 - \frac{2^5}{10} - 0$$

$$I_x = \frac{16}{4} - \frac{32}{10}$$

$$I_x = 4 - 3.2 = 0.8 \text{ m}^4$$

**10-22.** Determine the moment of inertia for the shaded area about the  $y$  axis.

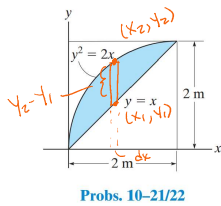


\* Parallel to  $y$ -axis  
vertical rectangle

$$I_y = \int x^2 dA$$

$$dA = (y_2 - y_1) dx$$

$$dA = ((2x)^{1/2} - x) dx$$



$$dA = (y_2 - y_1) dx$$

$$dA = ((2x)^{1/2} - x) dx$$

$$I_y = \int_0^2 x^2 ((2x)^{1/2} - x) dx$$

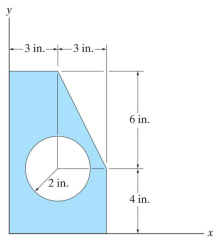
$$I_y = \int_0^2 \sqrt{2} x^{5/2} - x^3 dx$$

$$I_y = \sqrt{2} \cdot \frac{2}{7} x^{7/2} - \frac{1}{4} x^4 \bigg|_0^2$$

$$I_y = \frac{2\sqrt{2}}{7} (2)^{7/2} - \frac{1}{4} (2)^4 = \frac{2\sqrt{2}}{7} \cdot 8\sqrt{2} - \frac{1}{4} (16)$$

$$I_y = \frac{32}{7} - 4 = \frac{32}{7} - \frac{28}{7} = \frac{4}{7} \text{ m}^4$$

10-53. Determine the moment of inertia of the area about the y axis.



\* Parallel Axis Theorem

$$I_y = I_{y'} + A d^2$$

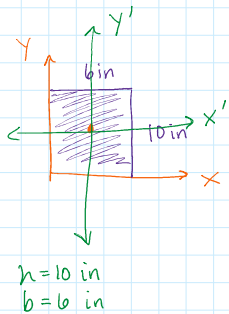
positive

$y$  &  $y'$  are parallel axis  
 $d$  - is shortest distance  
between two axis.

$$I_y > I_{y'}$$

Shaded area is further  
away from  $y$ -axis, than  
 $y'$ -axis.

$$I_y = (I_y)_{\text{rectangle}} - (I_y)_{\text{circle}} - (I_y)_{\text{triangle}}$$



$$I_y = \int_A x^2 dA \quad \text{or look up } I_y.$$

$$I_{y'} = \frac{1}{12} h b^3$$

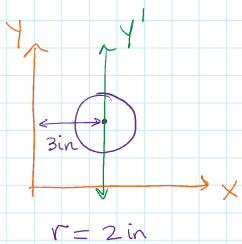
$$I_y = I_{y'} + A d^2$$

$$I_y = \frac{1}{12} h b^3 + (b \cdot h) \left(\frac{h}{2}\right)^2$$

$$I_y = \frac{1}{12} h b^3 + \frac{h}{4} b^3$$

$$I_y = \frac{1}{3} h b^3 = \frac{1}{3} (10 \text{ in}) (6 \text{ in})^3$$

$$I_y = 720 \text{ in}^4$$



$$I_y = \int_A x^2 dA \quad \text{or look up } I_y.$$

$$I_{y'} = \frac{1}{4} \pi r^4$$

$$I_y = I_{y'} + A d^2$$

$\frac{3}{2}$

$$I_y = \frac{1}{4} \pi r^4 + (\pi r^2) (1.5r)^2$$

$$I_y = \frac{1}{4} \pi r^4 + \frac{9}{4} \pi r^4$$

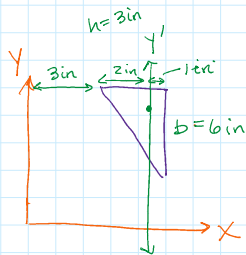
$$I_y = \frac{1}{4} \pi r^4 + (\pi r^2) (1/3 r)$$

$$I_y = \frac{1}{4} \pi r^4 + \frac{1}{3} \pi r^4$$

$$I_y = \frac{5}{2} \pi r^4 = \frac{5}{2} \pi (2)^4$$

$$I_y = \frac{5}{2} \pi (16) \text{ in}^4 = 40\pi \text{ in}^4$$

125.6



$$d = \frac{5}{3} h$$

$$h = 3 \text{ in}$$

$$d = \frac{5}{3} h$$

$$I_y = \int x^2 dA \quad \text{or look up } I_y$$

$$I_{y'} = \frac{1}{36} b h^3$$

$$I_y = I_{y'} + A d^2$$

$$I_y = \frac{1}{36} b h^3 + \left( \frac{1}{2} b h \right) \left( \frac{5}{3} h \right)^2$$

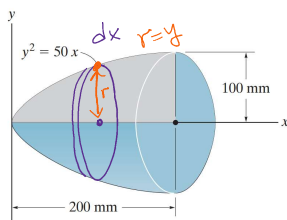
$$I_y = \frac{1}{36} b h^3 + \frac{25}{18} b h^3$$

$$I_y = \frac{51}{36} b h^3 = \frac{51}{36} (6) (3)^3$$

$$I_y = \frac{51 (27)}{6} = 229.5 \text{ in}^4$$

$$I_y = \overset{\text{rectangle}}{720} - \overset{\text{circle}}{40\pi} - \overset{\text{triangle}}{229.5} = 364.84 \text{ in}^4$$

10-86. Determine the radius of gyration  $k_x$  of the paraboloid. The density of the material is  $\rho = 5 \text{ Mg/m}^3$ .

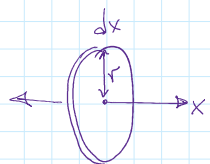


Prob. 10-86

$$k_x = \sqrt{\frac{I_x}{m}}$$

↓  
distance from x where you would lump the mass & get an equivalent mass moment of inertia about x-axis.

$$I_x = \checkmark \quad m = ?$$



look up mass moment of Inertia of an axis that goes through the center of disk.  
Tabulated @ end of textbook

$$dI_x = \frac{1}{2} \underline{dm} (\underline{r^2}) \rightarrow f(x, y, z)$$

$$dm = \rho dV = \rho \underbrace{\pi r^2 dx}_{dV}, \quad r = y \Rightarrow r^2 = y^2 = 50x$$

$$dI_x = \frac{1}{2} (50x) \rho \pi (50x) dx \rightarrow 1250 \rho \pi \int x^2 dx$$

$$dI_x = \frac{1}{2} (50x) \pi (50x) dx \rightarrow 1250 \pi \int_0^{200} x^2 dx$$

$$\int_0^{I_x} dI_x = \frac{(50)(50)(5 \times 10^3) \pi}{2} \int_0^{200 \text{ mm} = 0.2 \text{ m}} x^2 dx$$

$$\frac{1250 \pi}{3} (200)^3 = \frac{1250 \pi}{3} (8 \times 10^6)$$

$$I_x = \frac{1250 \pi}{3} (8 \times 10^6) = 10,666,666.67 \pi \text{ mm}^4$$

$$I_x = \frac{1250 \pi}{3} (8 \times 10^6) = 10,666,666.67 \pi \text{ mm}^4$$

$$dm = \pi r^2 dx = \pi (50x) dx$$

$$50 \pi \int_0^{200} x dx$$

$$\int_M dM = \pi (50) \int_0^{0.2 \text{ m}} x dx$$

$$M = \pi (50) \left[ \frac{1}{2} x^2 \right]_0^{0.2} = \pi (50) \left[ \frac{1}{2} (0.2)^2 - 0 \right]$$

$$M = (5 \times 10^3) \pi (25) (0.04) = 5000 \pi = 15,707.96 \text{ kg}$$

$$K_x = \sqrt{\frac{I_x}{M}} = \sqrt{\frac{10,666,666.67 \pi}{3 \cdot 5000 \pi}} = \sqrt{\frac{10}{3}} = 1.83 \text{ m}$$

$$\frac{50 \pi}{2} (200)^2$$

$$25 \pi (4 \times 10^4)$$

$$\frac{1250 \pi (8 \times 10^6)}{3 (25) \pi (4 \times 10^4)}$$

$$\sqrt{\frac{50}{3} \cdot 200}$$

$$\sqrt{\frac{10000}{3}} = 57.7 \text{ mm}$$

10-95. The slender rods have a mass of 4 kg/m. Determine the moment of inertia of the assembly about an axis perpendicular to the page and passing through point A.

$I_A$  is not tabulated for this shape.

Can look up moments of inertia for slender rods (section 1 & section 2) separately.

\* If the moment of inertia about axis of interest is not tabulated, then we can implement the parallel axis theorem.

$$I_x = I_{x'} + md^2$$

$$I_A = (I_A)_1 + (I_A)_2$$

$$(I_A)_1 = \frac{1}{12} m_1 l_1^2 - \text{from data sheet}$$

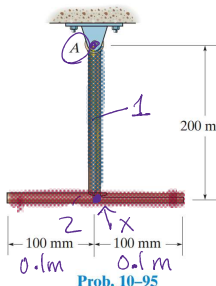
$$(I_A)_2 = \frac{1}{12} m_2 l_2^2 + m_2 d^2 - \text{from data sheet \& parallel-axis theorem}$$

$$M = 4 \text{ kg/m}$$

$$M_1 = 4 \text{ kg/m} (0.2 \text{ m})$$

$$m_1 = 0.8 \text{ kg}$$

$$200 \text{ mm} = 0.2 \text{ m}$$



$$l_2 = 0.2 \text{ m} \quad m_2 = 0.8 \text{ kg} \quad d = 0.2 \text{ m}$$

$$(I_A)_1 = \frac{1}{3} (0.8 \text{ kg}) (0.2 \text{ m})^2 = \frac{1}{3} (0.8) (0.04) = 0.01067 \text{ kg} \cdot \text{m}^2$$

$$(I_A)_2 = \frac{1}{12} (0.8 \text{ kg}) (0.2 \text{ m})^2 + (0.8 \text{ kg}) (0.2 \text{ m})^2$$

$$(0.8) (0.2)^2 \left[ \frac{1}{12} + 1 \right] = (0.8) (0.2)^2 \left( \frac{13}{12} \right) = 0.03467 \text{ kg} \cdot \text{m}^2$$

$$I_A = 0.01067 + 0.03467 = 0.04533 \text{ kg} \cdot \text{m}^2$$

$$0.04533 \text{ kg} \cdot \text{m}^2 \left( \frac{1000 \text{ mm}}{1 \text{ m}} \right) \left( \frac{1000 \text{ mm}}{1 \text{ m}} \right)$$

$$45330 \text{ kg} \cdot \text{mm}^2$$